Repair Complex CLR™ PF

A postbiotic approach towards photoaging

Repair Complex CLR™ PF

in vivo	ex vivo	in vitro
☑ skin	☑	☑

Repair Complex CLR™ PF is a biotechnological anti-aging active, obtained from a lysate of probiotic bifidobacteria, which potently protects against UV-induced damage in the skin. UV radiation leads to skin cell damage and immunosuppression, a deactivation of the local immune system in the skin.

By effectively activating anti-immunosuppressive processes on the cellular level, Repair Complex CLR™ PF increases cellular repair and potently supports the skin's endogenous protection mechanisms. Repair Complex CLR™ PF thus counteracts premature skin aging, i.e. photoaging.

Dosage: 5.0–10.0%
pH range: 3.8–7.0

INCI Name:
Bifida Ferment Lysate

For more formulations, click or scan QR-code

Skin benefits

- Postbiotic Bifida Ferment Lysate with a unique approach towards photoaging
- Strengthens the cellular immune system
- Allows the skin cells to effectively deal with UV light
- Skin cells maintain their functionality, allowing skin to keep its youthful appearance

Applications

- Skin protection

Marketing opportunities*

- Protects skin against premature aging
- Counteracts UV damage to maintain a radiant and youthful complexion
- Provides the skin with a comprehensive answer to aging
- Boosts the natural mechanism that extends the lifespan of youthful skin cells
- Activates natural cell protection processes

* This list is for illustrative purposes only.
Make sure to comply with relevant legislation.

RepairComplex CLR™ PF – Selected efficacy studies

In vitro (cell culture tests)

Protection against UV-induced immunosuppression

Depending on the dose, Repair Complex CLR™ PF reduces IL-10 release by more than 30% compared with control (Fig. 1).

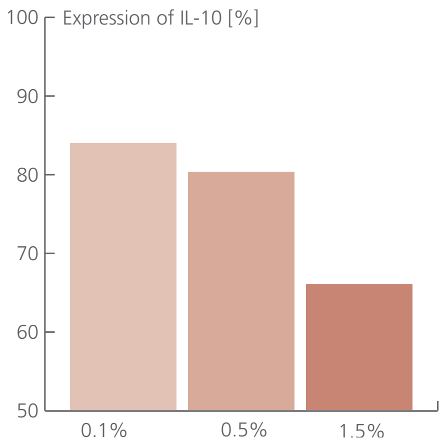

Fig. 1: Reduction of IL-10 expression after UV irradiation

Protection against DNA damage

Severe UV exposure leads to the formation of non-repairable cells with DNA double-strand breaks, which initiates apoptotic processes. Cells going into apoptosis show the formation of micronuclei. Repair Complex CLR™ PF significantly reduces the amount of cells going into apoptosis, clearly illustrating its ability to induce DNA repair (Fig. 2).

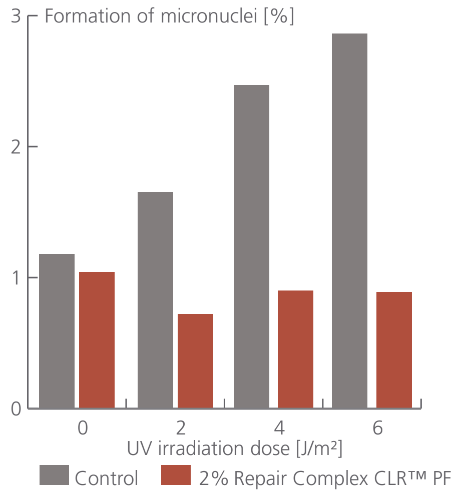

Fig. 2: Formation of micronuclei [%] after UV irradiation in normal fibroblast cells

In vivo (alternative test model)

Proof of DNA repair stimulation

The intact dorsal skin of 7 volunteers was treated on 5 test areas in the following manner:

A	O/W emulsion with **10% Repair Complex CLR™ PF**	UV-irradiated
B	Placebo	UV-irradiated
C	Untreated	UV-irradiated
D	Untreated	Non-irradiated
E	O/W emulsion with **10% Repair Complex CLR™ PF**	Non-irradiated

The area treated with Repair Complex CLR™ PF (A) showed an increase in DNA repair effectiveness in comparison with areas B and C. Without UV irradiation, no repair process was induced (D and E) (Fig. 3).

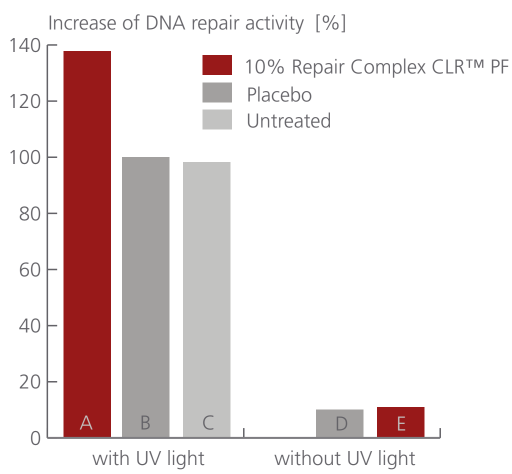

Fig. 3: Increase of DNA repair activity in vivo

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